The sealed statements of LinearAQFT

These 64 statements are the ones that `lake comparator` checks. Each is followed by the name of its formal version in `verify/Challenge.lean`.

Causal indices and nets

A causal index is the part of a spacetime that the axioms use: a preordered set R of regions, ordered by inclusion in the examples, with two relations, orthogonality $O_1 \perp O_2$ (spacelike separation) and causal dependence $O \triangleleft O'$ (O lies in the domain of dependence of O'). Points of spacetime do not appear. A C*-algebra is a unital complex C*-algebra, and a star subalgebra is a unital complex star subalgebra.

1. A causal index on a preordered set R is a pair of relations $\perp$ (orthogonality) and $\triangleleft$ (dependence) on R, and a pair of relations is a causal index exactly when $\perp$ is symmetric, $O_0 \le O_1$ and $O_1 \perp O_2$ imply $O_0 \perp O_2$, $O_0 \le O$ and $O \triangleleft O'$ imply $O_0 \triangleleft O'$, and $O \triangleleft O'$ and $O' \le O_1$ imply $O \triangleleft O_1$.
`Sealed.causalIndex_iff`

2. Isomorphisms of causal indices: for causal indices on R and R' and an order isomorphism $e : R \to R'$, there is an isomorphism of the causal indices acting on regions as e exactly when e preserves and reflects orthogonality and dependence: $e(O_1) \perp' e(O_2) \iff O_1 \perp O_2$ and $e(O) \triangleleft' e(O') \iff O \triangleleft O'$. An isomorphism is determined by its action on regions.
`Sealed.causalIndex_iso_iff`

3. The automorphisms of a causal index form a group in which the product is composition, $(\kappa_1\kappa_2)(O) = \kappa_1(\kappa_2(O))$, and the unit acts as the identity.
`Sealed.causalIndex_aut_mul`

4. A net over R in a C*-algebra A assigns to each region O a star subalgebra $\mathcal{A}(O)$ of A; an assignment $O \mapsto F(O)$ is a net exactly when it is monotone ($O \le O'$ implies $F(O) \subseteq F(O')$, isotony) and every $F(O)$ is norm-closed.
`Sealed.net_iff`

5. The axioms of a net $\mathcal{A}$ over a causal index on R. Generation: the norm closure of the star subalgebra generated by all local algebras is A. Causality: if $O_1 \perp O_2$, every element of $\mathcal{A}(O_1)$ commutes with every element of $\mathcal{A}(O_2)$. Time slice: if $O \triangleleft O'$, then $\mathcal{A}(O) \subseteq \mathcal{A}(O')$.
`Sealed.net_axioms_iff`

6. Covariance. Let $\mathcal{A}$ and $\mathcal{A}'$ be nets over causal indices on R and R' in C*-algebras A and A', κ an isomorphism of the causal indices and $\alpha : A \to A'$ a star isomorphism. Then (κ, α) is a covariance pair exactly when $\alpha(\mathcal{A}(O)) = \mathcal{A}'(\kappa(O))$ for every region O. Two nets $\mathcal{A}$, $\mathcal{B}$ over the same causal index are isomorphic exactly when some star isomorphism α of their algebras has $\alpha(\mathcal{A}(O)) = \mathcal{B}(O)$ for every O. For a group Γ with homomorphisms κ into the automorphisms of the causal index and α into the star automorphisms of A, the net is covariant under Γ exactly when $\alpha_\gamma(\mathcal{A}(O)) = \mathcal{A}(\kappa_\gamma(O))$ for every γ and O.

`Sealed.net_covariance_iff`

7. Nets generated by sets of elements. Let G assign to each region O a set $G(O)$ of elements of a C*-algebra A, monotonically in O, and let $\mathcal{A}(O)$ be the norm closure of the star subalgebra generated by $G(O)$; this is a net. If a commutes with b and with b^* whenever $a \in G(O_1)$, $b \in G(O_2)$ and $O_1 \perp O_2$, the net is causal. If $G(O) \subseteq \mathcal{A}(O')$ whenever $O \triangleleft O'$, it has the time slice property. If the union of all $G(O)$ generates A as a C*-algebra, the net is generating.
`Sealed.generated_net`

The CCR algebra

Let V be a real vector space and σ a real bilinear form on V. The Hilbert space $\ell^2(V)$ consists of the square-summable complex functions on the set V, and its bounded operators form a C*-algebra. A bilinear form is alternating when $\sigma(u,u) = 0$ for every u, and nondegenerate (separating) when $\sigma(u,w) = 0$ for all w implies $u = 0$.

8. The Weyl operators and the CCR algebra: for $u \in V$ the Weyl operator $W(u)$ is the bounded operator on $\ell^2(V)$ with

$$(W(u)f)(v) = e^{i\sigma(u,v)/2} f(v+u);$$

the CCR algebra $\mathrm{CCR}(V,\sigma)$ is the norm closure of the star subalgebra generated by the Weyl operators, a C*-algebra; and $W(u)$, regarded as an element of $\mathrm{CCR}(V,\sigma)$, is the same operator.
`Sealed.weyl_and_CCR`

9. For alternating σ every Weyl operator is unitary, $W(u)^* = W(-u)$, and the Weyl relation

$$W(u)W(w) = e^{-i\sigma(u,w)/2} W(u+w)$$

holds.
`Sealed.weyl_relations`

10. Representations. A representation π of a C*-algebra A on a complex Hilbert space H is a unital star homomorphism from A to the bounded operators on H. It is irreducible when every bounded operator commuting with all $\pi(a)$ is a scalar multiple of the identity. A has an irreducible representation when some irreducible representation on a nonzero Hilbert space exists, and A is primitive when it has an irreducible representation on a nonzero Hilbert space that is injective. (Formally, the Hilbert spaces are taken in the same type universe of the formal language as A.)
`Sealed.irreducible_primitive_iff`

11. Simplicity: for alternating nondegenerate σ the CCR algebra is a simple ring: its only two-sided ideals are 0 and the whole algebra, and these differ. If it moreover has an irreducible representation, it is primitive.
`Sealed.CCR_simple_primitive`

Phase-space data and the CCR net

Phase-space data over a causal index on R consist of a real vector space D of test functions, a real vector space V (the phase space) and the data of the next entry. For a test function f write $[f] \in V$ for its class. The Weyl operators $W(u)$ and the CCR algebra are those of the previous section, for the form σ of the phase-space data.

12. Phase-space data over a causal index on R, with test functions D and phase space V, consist of an assignment $O \mapsto D(O)$ of subspaces of D (the test functions localized in O), a bilinear form σ on V and a linear map $f \mapsto [f]$ from D to V; such a triple is phase-space data exactly when $O \mapsto D(O)$ is monotone, every test function lies in some $D(O)$, σ is alternating, $f \mapsto [f]$ is surjective, $\sigma([f],[g]) = 0$ whenever $f \in D(O_1)$, $g \in D(O_2)$ and $O_1 \perp O_2$ (causality), and whenever $O \triangleleft O'$ every $f \in D(O)$ has some $f' \in D(O')$ with $[f'] = [f]$ (time slice).
`Sealed.phaseSpace_iff`

13. Isomorphisms of phase-space data: for phase-space data Φ and Φ' over causal indices on R and R', an isomorphism κ of the causal indices and linear bijections $\theta : D \to D'$ and $T : V \to V'$, there is an isomorphism of the phase-space data made of κ, θ and T exactly when $\theta(D(O)) = D'(\kappa(O))$ for every region O, $[\theta f]' = T[f]$ for every test function f, and $\sigma'(Tu, Tw) = \sigma(u, w)$.
`Sealed.phaseSpace_iso_iff`

14. The CCR net of phase-space data: $\mathcal{A}(O)$ is the norm closure, in $\mathrm{CCR}(V, \sigma)$, of the star subalgebra generated by the Weyl operators $W([f])$ of the test functions $f \in D(O)$.
`Sealed.ccrNet_eq`

15. The axioms for phase-space data: for any phase-space data the CCR net is generating, causal and has the time slice property. If σ is nondegenerate and the CCR algebra has an irreducible representation, the CCR algebra is moreover primitive, which gives Axioms 1–4 of J. Dimock, Commun. Math. Phys. 77 (1980) 219–228.
`Sealed.ccrNet_axioms`

16. The Weyl operators of a region: for phase-space data and a region O, the local algebra $\mathcal{A}(O)$ is the norm closure of the complex linear span of the Weyl operators $W([f])$, $f \in D(O)$, and for $v \in V$ the Weyl operator $W(v)$ lies in $\mathcal{A}(O)$ exactly when $v = [f]$ for some $f \in D(O)$.
`Sealed.ccrNet_weyl`

17. Covariance of the CCR net. For an isomorphism θ of phase-space data $\Phi \to \Phi'$ with map T of phase spaces there is a star isomorphism $\alpha_\theta : \mathrm{CCR}(V, \sigma) \to \mathrm{CCR}(V', \sigma')$ with $\alpha_\theta(W(u)) = W(Tu)$ for every u (a star isomorphism is determined by these values); $(\kappa_\theta, \alpha_\theta)$, with κ_θ the isomorphism of causal indices of θ, is a covariance pair of the CCR nets; α of the identity isomorphism is the identity; and

$$\alpha_{\theta_2 \circ \theta_1} = \alpha_{\theta_2} \circ \alpha_{\theta_1}.$$

`Sealed.ccrNet_covariance`

18. Group covariance. Let Γ be a group and θ_γ ($\gamma \in \Gamma$) isomorphisms of phase-space data Φ onto itself, not assumed to compose, whose maps T_γ of the phase space satisfy $T_1 = \mathrm{id}$ and $T_{\gamma\gamma'} = T_\gamma T_{\gamma'}$, and let κ be a homomorphism from Γ to the automorphisms of the causal index with κ_γ the isomorphism of causal indices of θ_γ. Then there is a homomorphism α from Γ to the star automorphisms of the CCR algebra with $\alpha_\gamma(W(u)) = W(T_\gamma u)$, and the CCR net is covariant under Γ through κ and α.
`Sealed.ccrNet_isCovariantUnder`

19. Weyl systems: for an alternating form σ on V and a C*-algebra A, a Weyl system is a map $W : V \to A$; a map is a Weyl system exactly when every $W(u)$ is unitary,

$$W(u)W(w) = e^{-i\sigma(u,w)/2} W(u + w),$$

and the $W(u)$ generate A as a C*-algebra. The Weyl system of the CCR algebra is the map $u \mapsto W(u)$ into $\mathrm{CCR}(V, \sigma)$.

`Sealed.weylSystem_iff`

20. The net of a Weyl system: for phase-space data and a Weyl system W for its form in a C*-algebra A, let $\mathcal{A}_W(O)$ be the norm closure of the star subalgebra of A generated by the $W([f])$, $f \in D(O)$. This net is generating, causal and has the time slice property. If W' is a second Weyl system for the same form, in A', and $\alpha : A \to A'$ is a star isomorphism with $\alpha(W(u)) = W'(u)$ for every u, the two nets are isomorphic.
`Sealed.weylNet_axioms`

Green data

Green data describe a linear field equation $Pu = 0$ through its retarded and advanced Green operators $G^{\pm}$. They consist of a set X of points with a relation $p \preceq q$ (q in the causal future of p), a real vector space $\mathcal{F}$ of functions on X with a support map, a real vector space D of test functions mapped linearly into $\mathcal{F}$ by $f \mapsto \bar{f}$, the operator P, the pairing, the Green operators and a family of cutoffs, as in the next entry. Two points are comparable when $p \preceq q$ or $q \preceq p$, and $J(K)$ is the set of points comparable with a point of K. The commutator function is $E = G^- - G^+$ (advanced minus retarded). Causal dependence of regions is defined through the cutoffs; if Green data have no cutoffs, no region depends on any region and the time slice property says nothing. The Green data of the lattice and of Minkowski space have cutoffs, given in their sections.

21. Green data on X with test functions D and functions $\mathcal{F}$, a family of cutoffs $(\chi_c, S_c)_{c \in K}$ indexed by a type K (χ_c a linear map of $\mathcal{F}$, S_c a subset of X, the slab), and the remaining data as named below: there are Green data with these data, whose cutoff pairs (χ, S) are exactly the pairs (χ_c, S_c), if and only if the support of 0 is empty,

$$\operatorname{supp}(h + h') \subseteq \operatorname{supp} h \cup \operatorname{supp} h'$$

and $\operatorname{supp}(ch) \subseteq \operatorname{supp} h$; P_D is a linear map of D with $\overline{P_D f} = P\bar{f}$; the bilinear pairing $\langle f, h \rangle$ of D with $\mathcal{F}$ vanishes when $\operatorname{supp} \bar{f}$ and $\operatorname{supp} h$ are disjoint, and $h = 0$ when $\langle f, h \rangle = 0$ for all f; the linear maps $G^{\pm} : D \to \mathcal{F}$ satisfy $PG^{\pm} f = \bar{f}$, $G^{\pm} P_D f = \bar{f}$,

$$\operatorname{supp} G^+ f \subseteq \{q : p \preceq q \text{ for some } p \in \operatorname{supp} \bar{f}\},$$

$$\operatorname{supp} G^- f \subseteq \{q : q \preceq p \text{ for some } p \in \operatorname{supp} \bar{f}\}$$

and $\langle f, G^+ g \rangle = \langle g, G^- f \rangle$; and for every cutoff c and test function f,

$$\operatorname{supp} P(\chi_c E f) \subseteq S_c \cap J(\operatorname{supp} \bar{f})$$

and

$$(G^+ f - \chi_c G^+ f) + \chi_c G^- f = \bar{f}_c$$

for some test function f_c.
`Sealed.greenData_iff`

22. The causal index and the phase-space data of Green data. Regions are subsets of X: $O_1 \perp O_2$ when no point of O_1 is comparable with a point of O_2, and $O \triangleleft O'$ when for every test function f with $\operatorname{supp} \bar{f} \subseteq O$ some cutoff has $S_c \cap J(\operatorname{supp} \bar{f}) \subseteq O'$. The test functions localized in O are those with $\operatorname{supp} \bar{f} \subseteq O$; the phase space is the quotient $D / \ker E$ with $E = G^- - G^+$, the class $[f]$ is the coset of f, and the form is $\sigma([f], [g]) = \langle f, Eg \rangle$.
`Sealed.greenData_phaseSpace`

23. The axioms for Green data: for all Green data the form σ is nondegenerate, and the CCR net of the derived phase-space data is generating, causal and has the time slice property; if the CCR algebra has an irreducible representation, it is moreover primitive.
`Sealed.greenData_axioms`

24. Symmetries of Green data: a symmetry consists of a bijection γ of X, linear bijections γ_D of D and $\gamma_{\mathcal{F}}$ of $\mathcal{F}$ and a bijection γ_K of the cutoffs; such a quadruple is a symmetry exactly when γ preserves and reflects $\preceq$, $\overline{\gamma_D f} = \gamma_{\mathcal{F}} \bar{f}$,

$$\operatorname{supp} \gamma_{\mathcal{F}} h = \gamma(\operatorname{supp} h),$$

$G^{\pm} \gamma_D f = \gamma_{\mathcal{F}} G^{\pm} f$,

$$\langle \gamma_D f, \gamma_{\mathcal{F}} h \rangle = \langle f, h \rangle$$

and $S_{\gamma_K c} = \gamma(S_c)$.
`Sealed.greenData_symmetry_iff`

25. Covariance from symmetries. Let Γ be a group and s_γ ($\gamma \in \Gamma$) symmetries of Green data, not assumed to compose, whose maps of test functions satisfy $(s_1)_D = \mathrm{id}$ and

$$(s_{\gamma\gamma'})_D = (s_\gamma)_D (s_{\gamma'})_D,$$

and let κ be a homomorphism from Γ to the automorphisms of the derived causal index with $\kappa_\gamma(O) = s_\gamma(O)$, the image of O under the bijection of points. Then there is a homomorphism α from Γ to the star automorphisms of the CCR algebra with $\alpha_\gamma(W([f])) = W([(s_\gamma)_D f])$, and the CCR net is covariant under Γ through κ and α.
`Sealed.greenData_isCovariantUnder`

The lattice causal index

A point of the lattice $\mathbb{Z}^2$ is a pair $p = (t, x)$, time first; we write $p_t = t$ and $p_x = x$. Regions are all subsets of $\mathbb{Z}^2$, ordered by inclusion. Two points are comparable when one lies in the causal future of the other.

26. The lattice causal order and causal index. A point of the lattice is a pair of integers, and $(t, x) \leq (t', x')$ if and only if $|x' - x| \leq t' - t$. The causal shadow of a set O of points is

$$J(O) = \{p : p \leq q \text{ or } q \leq p \text{ for some } q \in O\},$$

and the pair of consecutive time slices at $t_1 \in \mathbb{Z}$ is

$$S_{t_1} = \{p : p_t = t_1 \text{ or } p_t = t_1 + 1\}.$$

In the lattice causal index, two regions O_1, O_2 are orthogonal if and only if no point of O_1 is comparable with a point of O_2, and O depends on O' if and only if for every finite $Y \subseteq O$ there is $t_1 \in \mathbb{Z}$ with $J(Y) \cap S_{t_1} \subseteq O'$.
`Sealed.lattice_causal_index`

The discrete Klein–Gordon equation

A test function is a finitely supported function $f : \mathbb{Z}^2 \to \mathbb{R}$, with support $\operatorname{supp} f = \{p : f(p) \neq 0\}$, a finite set; δ_p is the test function with value 1 at p and 0 elsewhere. The mass parameter m^2 is an arbitrary real number, of either sign.

27. The discrete Klein–Gordon operator and its formal self-adjointness. Test functions are the finitely supported functions $\mathbb{Z}^2 \to \mathbb{R}$. For $m^2 \in \mathbb{R}$ and $h : \mathbb{Z}^2 \to \mathbb{R}$,

$$(Ph)(t,x) = h(t+1,x) + h(t-1,x) - h(t,x+1) - h(t,x-1) + m^2 h(t,x);$$

P maps every test function f to a test function whose values are those of Pf. The pairing of a test function f with h is

$$\langle f, h\rangle = \sum_{p \in \operatorname{supp} f} f(p)h(p),$$

and $\langle Pf, h\rangle = \langle f, Ph\rangle$ for every test function f and every $h : \mathbb{Z}^2 \to \mathbb{R}$.
`Sealed.lattice_kg_operator`

28. Uniqueness: let $m^2 \in \mathbb{R}$ and let $h : \mathbb{Z}^2 \to \mathbb{R}$ satisfy $Ph = 0$ for the discrete Klein–Gordon operator P. If there is $t_0 \in \mathbb{Z}$ with $h(t,x) = 0$ for all $t < t_0$ and all x, or one with $h(t,x) = 0$ for all $t > t_0$ and all x, then $h = 0$.
`Sealed.lattice_uniqueness`

Green operators and the commutator function

29. The retarded and advanced Green operators. Let $m^2 \in \mathbb{R}$, P the discrete Klein–Gordon operator and f a test function. The retarded and advanced Green operators satisfy $PG^+f = PG^-f = f$ and $G^+Pf = G^-Pf = f$. If $G^+f(p) \neq 0$, there is $q \in \operatorname{supp} f$ with $|p_x - q_x| \leq p_t - q_t - 1$; if $G^-f(p) \neq 0$, there is $q \in \operatorname{supp} f$ with $|p_x - q_x| \leq q_t - p_t - 1$ (strictly inside the future, respectively past, cone of the support). These determine the operators: a function $h : \mathbb{Z}^2 \to \mathbb{R}$ equals G^+f if and only if $Ph = f$ and h vanishes at all times $t < t_0$ for some $t_0 \in \mathbb{Z}$, and it equals G^-f if and only if $Ph = f$ and h vanishes at all times $t > t_0$ for some $t_0 \in \mathbb{Z}$.
`Sealed.lattice_green_operators`

30. The commutator function. Let $m^2 \in \mathbb{R}$, $G^{\pm}$ the Green operators and P the discrete Klein–Gordon operator. The causal propagator is $Ef = G^-f - G^+f$ (advanced minus retarded) for every test function f. For all test functions f, g: $PEf = 0$ and $EPf = 0$; $\langle f, Eg\rangle = -\langle g, Ef\rangle$; and if $\operatorname{supp} f$ and $\operatorname{supp} g$ are orthogonal in the lattice causal index, then $\langle f, Eg\rangle = 0$. Moreover $\langle \delta_{(0,x)}, E\delta_{(1,y)}\rangle$ is 1 if $x = y$ and 0 otherwise, for all $x, y \in \mathbb{Z}$.
`Sealed.lattice_commutator`

Time slice

31. The time slice. Let $m^2 \in \mathbb{R}$ and E the causal propagator. For every test function f and $t_1 \in \mathbb{Z}$ there is a test function f' supported in $S_{t_1} \cap J(\operatorname{supp} f)$, with

$$f'(t_1, x) = -(Ef)(t_1 + 1, x)$$

and

$$f'(t_1 + 1, x) = (Ef)(t_1, x)$$

for every x, and with $Ef' = Ef$. Consequently, if O depends on O' in the lattice causal index and $\operatorname{supp} f \subseteq O$, there is a test function f' with $\operatorname{supp} f' \subseteq O'$ and $Ef' = Ef$.
`Sealed.lattice_time_slice`

The lattice phase-space data

32. The lattice phase-space data and nondegeneracy. Let $m^2 \in \mathbb{R}$ and E the causal propagator, a linear map on test functions. In the lattice phase-space data, the test functions localized in a region O are those with $\operatorname{supp} f \subseteq O$; the phase space is the quotient V of the test functions by the kernel of E, and the class map sends f to its class $[f]$, so that $[f] = [g]$ if and only if $Ef = Eg$; and the form is $\sigma([f],[g]) = \langle f, Eg\rangle$. Moreover $\sigma([f],[\delta_p]) = -(Ef)(p)$ for every test function f and point p, and σ is nondegenerate: if $\sigma(u,w) = 0$ for every $w \in V$, then $u = 0$.

`Sealed.lattice_phase_space`

The lattice CCR net

33. Translations of the lattice. Let $m^2 \in \mathbb{R}$. The translation of the lattice causal index by $a \in \mathbb{Z}^2$ maps a region O to $O + a = \{p : p - a \in O\}$, and the translation $a \mapsto \alpha_a$ of the CCR algebra of the lattice phase-space data maps the Weyl operator of $[f]$ to that of the translated test function: if $g(p) = f(p - a)$ for every point p, then $\alpha_a(W([f])) = W([g])$. Both maps are group homomorphisms from the additive group $\mathbb{Z}^2$, into the automorphisms of the causal index and the star automorphisms of the CCR algebra respectively.

`Sealed.lattice_translations`

34. The net of the lattice field. For every $m^2 \in \mathbb{R}$, with no further hypothesis, the CCR net of the lattice phase-space data is generating, causal for the lattice causal index, has the time-slice property for it, and is covariant under the translations of $\mathbb{Z}^2$, acting on regions and on the CCR algebra as in the statement on translations; and the CCR algebra $\mathrm{CCR}(V,\sigma)$ of the lattice phase space is a simple ring. If $\mathrm{CCR}(V,\sigma)$ has an irreducible representation, it is primitive. (Generation, primitivity, causality and time slice are Axioms 1 to 4 of Dimock, 1980.)

`Sealed.lattice_ccrNet_axioms`

The Cauchy-data net

Cauchy data are pairs $d = (d_0, d_1)$ of finitely supported functions $\mathbb{Z} \to \mathbb{R}$, the values of a solution on two consecutive time slices.

35. The Cauchy map. Let $m^2 \in \mathbb{R}$, E the causal propagator and (V,σ) the lattice phase space. Cauchy data are pairs $d = (d_0, d_1)$ of finitely supported functions $\mathbb{Z} \to \mathbb{R}$, with the Wronskian

$$\sigma_S(d, d') = \sum_x d_0(x) d_1'(x) - \sum_x d_1(x) d_0'(x)$$

(finite sums over the supports of d_0 and d_1). For $t_1 \in \mathbb{Z}$, the Cauchy map Λ_{t_1} sends $[f]$ to $((Ef)(t_1,\cdot),(Ef)(t_1+1,\cdot))$; it is a linear bijection from V onto the Cauchy data, and

$$\sigma_S(\Lambda_{t_1} u, \Lambda_{t_1} w) = \sigma(u,w)$$

for all $u, w \in V$.

`Sealed.lattice_cauchy_map`

36. The Cauchy-data net. Let $m^2 \in \mathbb{R}$, $t_1 \in \mathbb{Z}$, Λ_{t_1} the Cauchy map and W_S the Weyl operators of the CCR algebra $\mathrm{CCR}(V_S,\sigma_S)$ of the Cauchy data with the Wronskian. The Cauchy-data net assigns to a region O the norm closure of the star subalgebra of $\mathrm{CCR}(V_S,\sigma_S)$ generated by the $W_S(\Lambda_{t_1}[f])$ for the test functions f with $\operatorname{supp} f \subseteq O$. It is isomorphic to the CCR net of the lattice phase-space data: a star isomorphism of the CCR algebras maps each local algebra onto the local algebra of the same region.

`Sealed.lattice_cauchyNet_isomorphic`

37. The leapfrog step. Let $m^2 \in \mathbb{R}$ and let F map Cauchy data (d_0, d_1) to $(d_1, -d_0 + d_1(\cdot + 1) + d_1(\cdot - 1) - m^2 d_1)$. Then $\Lambda_{t_1+1} = F \circ \Lambda_{t_1}$ for the Cauchy maps at every $t_1 \in \mathbb{Z}$, and F preserves the Wronskian: $\sigma_S(Fd, Fd') = \sigma_S(d, d')$.

`Sealed.lattice_leapfrog`

The lattice as Green data

38. The lattice field as Green data. Let $m^2 \in \mathbb{R}$. The lattice Green data have the points of $\mathbb{Z}^2$ with the lattice causal order, the functions $\mathbb{Z}^2 \to \mathbb{R}$ with their supports, the test functions included as functions, the discrete Klein–Gordon operator P on functions and on test functions, the pairing $\langle f, h \rangle$, the retarded and advanced Green operators G^+ and G^-, and cutoffs indexed by $t_1 \in \mathbb{Z}$, the cutoff χ_{t_1} keeping a function at the times $t \geq t_1 + 1$ and setting it to 0 before, with slab S_{t_1}. Their derived causal index is the lattice causal index, their propagator is E, their test functions localized in O are those of the lattice phase-space data, and there is an isomorphism from their phase-space data to the lattice phase-space data that is the identity on regions and on test functions and maps the class of f to $[f]$.

`Sealed.lattice_greenData`

Minkowski space, test functions and the wave operator

Minkowski space is $M = \mathbb{R} \times \mathbb{R}^3$, a point $p = (p^0, \vec{p})$ with time p^0 and position $\vec{p}$; $|\cdot|$ is the Euclidean norm on $\mathbb{R}^3$ and $\vec{u} \cdot \vec{w}$ its inner product. The Minkowski form is

$$\eta(u, w) = -u^0 w^0 + \vec{u} \cdot \vec{w}$$

(signature mostly plus) and $e_0 = (1, \vec{0})$. Integrals over M and over $\mathbb{R}^3$ are with respect to Lebesgue measure. A test function is a smooth real function on M with compact support, a smooth function is a real function on M that is differentiable infinitely often.

39. Spacetime and the causal order: the points of M are the pairs $(p^0, \vec{p})$ with $p^0 \in \mathbb{R}$ and $\vec{p}$ in $\mathbb{R}^3$ with the Euclidean norm. The point q lies in the closed causal future of p, written $p \leq q$, if and only if $|\vec{q} - \vec{p}| \leq q^0 - p^0$. For a set K of points, the causal future $J^+(K)$ is the set of points q with $p \leq q$ for some $p \in K$, and the causal past $J^-(K)$ the set of points q with $q \leq p$ for some $p \in K$.

`Sealed.minkowski_point_causalLE`

40. Test functions and smooth functions: the test functions $\mathcal{D} = C_c^\infty(M)$ are the elements of the standard space of smooth compactly supported functions on the open set M; each test function is a real function on M, distinct test functions are distinct functions, and the functions so obtained are exactly the smooth functions with compact support. The smooth functions $\mathcal{F} = C^\infty(M)$ are the real functions on M that are differentiable infinitely often, a real vector space of functions under pointwise operations, and the inclusion $\iota : \mathcal{D} \to \mathcal{F}$ sends a test function to itself.

`Sealed.minkowski_testFun_fn`

41. Supports and the pairing: the support of a smooth function h is the closure

$$\operatorname{supp} h = \overline{\{p : h(p) \neq 0\}}.$$

For a test function f and a smooth function h, the product fh is integrable on M and the pairing is

$$\langle f, h \rangle = \int_M f(p) h(p)\, dp.$$

`Sealed.minkowski_supp_pairing`

42. The wave operator

$$P = \partial_0^2 - \sum_{i=1}^{3} \partial_i^2 :$$

write

$$\partial_v u(p) = \frac{d}{ds} u(p + sv)\big|_{s=0}$$

for the derivative of a function u at p along the vector v, $e_0 = (1, \vec{0})$ and $e_i = (0, \vec{e}_i)$ with $\vec{e}_i$ the standard basis of $\mathbb{R}^3$. For every smooth function h and point p,

$$(Ph)(p) = \partial_{e_0}(\partial_{e_0} h)(p) - \sum_{i=1}^{3} \partial_{e_i}(\partial_{e_i} h)(p),$$

and P maps test functions to test functions: $\iota(P_{\mathcal{D}} f) = P(\iota f)$ for every test function f, where $P_{\mathcal{D}}$ is the wave operator on test functions.

`Sealed.minkowski_wave`

The Kirchhoff operators

The retarded and advanced Green operators $G^{\pm}$ of the wave operator are given by the Kirchhoff formula. G^+ samples a test function on the backward light cone of the point, so $G^+ f$ is supported in the causal future of $\operatorname{supp} f$.

43. The Kirchhoff formula: for every test function f and point $p = (t, x)$, the functions $y \mapsto f(t \mp |y|, x - y)/|y|$ on $\mathbb{R}^3$ are integrable, and the retarded and advanced Green operators are

$$(G^{\pm} f)(t, x) = \frac{1}{4\pi} \int_{\mathbb{R}^3} f(t \mp |y|, x - y)\, |y|^{-1}\, dy.$$

`Sealed.minkowski_kirchhoff`

44. The Kirchhoff operators are retarded and advanced Green operators of the wave operator P: for all test functions f and g, $PG^{\pm} f = f$ and $G^{\pm} P f = f$; the support of $G^+ f$ lies in $J^+(\operatorname{supp} f)$ and that of $G^- f$ in $J^-(\operatorname{supp} f)$; and $\langle f, G^+ g \rangle = \langle g, G^- f \rangle$.

`Sealed.minkowski_green_operators`

45. Uniqueness: let G' be any map (not assumed linear) from test functions to smooth functions with $PG' f = f$ for every test function f. If

$$\operatorname{supp} G' f \subseteq J^+(\operatorname{supp} f)$$

for every f, then $G' = G^+$; if

$$\operatorname{supp} G' f \subseteq J^-(\operatorname{supp} f)$$

for every f, then $G' = G^-$.

`Sealed.minkowski_green_unique`

The free massless field as Green data

The Green data of the wave operator consist of the objects above together with the inertial slabs as cutoffs. The time function of a vector n is

$$\tau_n(p) = -\eta(n, p) = n^0 p^0 - \vec{n} \cdot \vec{p},$$

and n is future unit when $n^0 > 0$ and $\eta(n, n) = -1$, that is $(n^0)^2 - |\vec{n}|^2 = 1$. The smooth transition function is the standard $s(x) = e(x)/(e(x) + e(1 - x))$ with $e(x) = \exp(-1/x)$ for $x > 0$ and $e(x) = 0$ for $x \leq 0$; it is smooth, 0 for $x \leq 0$ and 1 for $x \geq 1$.

46. The fields of the Green data of the wave operator: the causal relation is $\leq$, the support map is supp, the map of test functions to functions is ι, the operator is the wave operator P on smooth functions and $P_{\mathcal{D}}$ on test functions, the retarded and advanced Green operators are the Kirchhoff operators G^+ and G^-, and the pairing is

$$\langle f, h\rangle = \int_M fh.$$

`Sealed.minkowski_greenData_fields`

47. The cutoffs of the Green data of the wave operator are the inertial slabs: they correspond bijectively to the triples (n, t_1, t_2) of a vector $n \in M$ with $n^0 > 0$ and $(n^0)^2 - |\vec{n}|^2 = 1$ and real numbers $t_1 < t_2$. For the cutoff c with triple (n, t_1, t_2), the slab of c is $\{p : t_1 \leq \tau_n(p) \leq t_2\}$ with $\tau_n(p) = n^0 p^0 - \vec{n}\cdot\vec{p}$, and the cutoff map χ_c multiplies a smooth function h by $s((\tau_n - t_1)/(t_2 - t_1))$:

$$(\chi_c h)(p) = s\big((\tau_n(p) - t_1)/(t_2 - t_1)\big)\, h(p),$$

with s the smooth transition function.

`Sealed.minkowski_greenData_cutoffs`

48. The axioms of Dimock (1980) for the free massless field: the CCR net of the Green data of the wave operator, over its derived causal index, is generating, causal and has the time-slice property, with no hypothesis; and if the CCR algebra of its phase space has an irreducible representation, the CCR algebra is primitive.

`Sealed.minkowski_ccrNet_axioms`

Poincaré covariance

The orthochronous Poincaré group consists of the maps $p \mapsto Lp + a$ with L a linear map of M preserving η and with $(Le_0)^0 > 0$, and $a \in M$. No condition is placed on the determinant of L, so space reflections such as $(t, x) \mapsto (t, -x)$ belong to the group.

49. The orthochronous Poincaré group: each element g acts on points by a map $p \mapsto gp$; the map $g \mapsto (p \mapsto gp)$ is injective and turns the group product into composition, $(gg')p = g(g'p)$; and its image is exactly the set of maps $p \mapsto Lp + a$ with $L : M \to M$ linear, $\eta(Lu, Lw) = \eta(u, w)$ for all u, w, $(Le_0)^0 > 0$, and $a \in M$.

`Sealed.minkowski_poincare_group`

50. Poincaré covariance of the free massless field: the action of an orthochronous Poincaré map g on regions sends O to its image $g(O)$; its action α_g on the CCR algebra of the phase space of the Green data of the wave operator sends the Weyl operator $W([f])$ of a test function f to $W([f'])$ for every test function f' with $f'(gp) = f(p)$ for all p (that is, $f' = f \circ g^{-1}$); and the CCR net is covariant under the orthochronous Poincaré group with these two actions.

`Sealed.minkowski_poincare_covariance`

The Klein–Gordon field

For every real μ (the case $\mu = m^2$ with $m \geq 0$ is the free field of mass m; $\mu < 0$ is allowed), the Klein–Gordon operator is $P_\mu = P + \mu$, with P the wave operator. Its Green operators are the series

$$G^\pm_\mu f = \sum_{k\geq 0} (-\mu)^k (K^\pm)^{k+1} f,$$

with $K^\pm$ the Kirchhoff integrals applied to arbitrary functions (the series entry below); for $k \geq 1$ the functions $(K^\pm)^k f$ are not compactly supported.

51. The Klein–Gordon operator: for every real μ, every smooth function h and point p,

$$(P_\mu h)(p) = (Ph)(p) + \mu\, h(p),$$

and $\iota(P_{\mu,\mathcal{D}} f) = P_\mu(\iota f)$ for every test function f, where $P_{\mu,\mathcal{D}}$ is the Klein–Gordon operator on test functions.

`Sealed.kg_operator`

52. The Green operators of the Klein–Gordon operator as a series: let $K^\pm$ be the maps on all real functions h on M given by

$$(K^\pm h)(t,x) = \frac{1}{4\pi} \int_{\mathbb{R}^3} h(t \mp |y|, x - y)\, |y|^{-1}\, dy$$

(the Kirchhoff integrals, applied to functions that need not have compact support; the integral is taken as 0 when the integrand is not integrable, and the statement proves integrability wherever $K^\pm$ is applied below), and $(K^\pm)^k$ their k-th iterates, $(K^\pm)^0 f = f$. For every real μ and test function f: for every $k \geq 0$ and point (t,x) the integrands $y \mapsto ((K^\pm)^k f)(t \mp |y|, x - y)/|y|$ are integrable on $\mathbb{R}^3$; and for every point p the series

$$\sum_{k \geq 0} (-\mu)^k ((K^\pm)^{k+1} f)(p)$$

converge (unconditionally, which for real series is absolute convergence) with sums $(G^+_\mu f)(p)$ and $(G^-_\mu f)(p)$ respectively.

`Sealed.kg_series`

53. Green operators of the Klein–Gordon operator: for every real μ and all test functions f and g, $P_\mu G^\pm_\mu f = f$ and $G^\pm_\mu P_\mu f = f$; the support of $G^+_\mu f$ lies in $J^+(\operatorname{supp} f)$ and that of $G^-_\mu f$ in $J^-(\operatorname{supp} f)$; and

$$\langle f, G^+_\mu g \rangle = \langle g, G^-_\mu f \rangle.$$

`Sealed.kg_green_operators`

54. Uniqueness for the Klein–Gordon operator: let μ be real and G' any map (not assumed linear) from test functions to smooth functions with $P_\mu G' f = f$ for every test function f. If

$$\operatorname{supp} G' f \subseteq J^+(\operatorname{supp} f)$$

for every f, then $G' = G^+_\mu$; if

$$\operatorname{supp} G' f \subseteq J^-(\operatorname{supp} f)$$

for every f, then $G' = G^-_\mu$.

`Sealed.kg_green_unique`

55. The fields of the Green data of the Klein–Gordon operator, for every real μ: the causal relation, the support map, the map ι and the pairing are those of Minkowski space; the operator is P_μ on smooth functions and on test functions; the retarded and advanced Green operators are G^+_μ and G^-_μ; and the cutoffs, with their slabs and cutoff maps, are those of the Green data of the wave operator: there is a bijection e from the cutoffs of the Klein–Gordon Green data to those of the wave operator's with the same slab and the same cutoff map, $S_c = S_{e(c)}$ and $\chi_c = \chi_{e(c)}$.

`Sealed.kg_greenData_fields`

56. The axioms of Dimock (1980) for the Klein–Gordon field, for every real μ: the CCR net of the Green data of P_μ, over its derived causal index, is generating, causal and has the time-slice property, with no hypothesis; and if the CCR algebra of its phase space has an irreducible representation, the CCR algebra is primitive.
`Sealed.kg_ccrNet_axioms`

57. Poincaré covariance of the Klein–Gordon field, for every real μ: P_μ with its Green operators $G_\mu^\pm$ forms a family of Poincaré-invariant Green operators whose Green data are the Green data of P_μ, and the actions below are those this family defines. The action of an orthochronous Poincaré map g on regions sends O to $g(O)$; its action α_g on the CCR algebra of the phase space of the Green data of P_μ sends the Weyl operator $W([f])$ of a test function f to $W([f'])$ for every test function f' with $f'(gp) = f(p)$ for all p; and the CCR net is covariant under the orthochronous Poincaré group with these two actions.
`Sealed.kg_poincare_covariance`

58. The massless field is the case $\mu = 0$: $P_0 = P$ on smooth functions and on test functions, $G_0^+ = G^+$, $G_0^- = G^-$, and the Green data of P_0 are the Green data of the wave operator.
`Sealed.kg_zero`

Huygens' principle

The future light-cone surface is

$$\partial C^+ = \{v \in M : v^0 = |\vec{v}|\}.$$

A point p lies in the open future cone of q when $|\vec{p} - \vec{q}| < p^0 - q^0$.

59. Huygens' principle: for the wave operator the Green operators propagate only along the light cone,

$$\operatorname{supp} G^+ f \subseteq \operatorname{supp} f + \partial C^+$$

and

$$\operatorname{supp} G^- f \subseteq \operatorname{supp} f - \partial C^+$$

for every test function f (sums of sets taken elementwise). For the Klein–Gordon operator with real μ: one has $(G_\mu^+ f)(p) = 0$ for every test function f and every point p lying in the open future cone of every point of $\operatorname{supp} f$ if and only if $\mu = 0$; and one has $(G_\mu^- f)(p) = 0$ for every test function f and every point p such that every point of $\operatorname{supp} f$ lies in the open future cone of p if and only if $\mu = 0$.
`Sealed.kg_huygens`

Examples

These statements show that the hypotheses of the theorems above can be met and that the objects are not degenerate. No example of an irreducible representation of a CCR algebra is proved; the statements that assume one are conditional on it.

60. The form is nonzero: for every $m^2 \in \mathbb{R}$, in the lattice phase space,

$$\sigma([\delta_{(0,0)}], [\delta_{(3,0)}]) = 1 + (m^2)^2 = 1 + m^4.$$

`Sealed.witness_lattice_sigma`

61. Causality is not vacuous: for every $m^2 \in \mathbb{R}$ there are regions O_1, O_2, orthogonal in the lattice causal index, and test functions f localized in O_1 and g localized in O_2 whose classes $[f]$, $[g]$ in the lattice phase space are nonzero.
`Sealed.witness_lattice_orthogonal`

62. The time-slice property is not isotony: there are regions O, O' of the lattice such that O depends on O' in the lattice causal index and $O \not\subseteq O'$.
`Sealed.witness_lattice_dependent_not_subset`

63. Example: for every vector $n \in M$ with $n^0 > 0$ and $(n^0)^2 - |\vec{n}|^2 = 1$ and all real $t_1 < t_2$, the Green data of the wave operator have a cutoff whose slab is

$$\{p : t_1 \le n^0 p^0 - \vec{n} \cdot \vec{p} \le t_2\}.$$

`Sealed.witness_minkowski_cutoff`

64. Example (failure of Huygens' principle with a mass term): for every real $\mu \neq 0$ there are a test function f and a point p in the open future cone of every point of $\operatorname{supp} f$ with $(G^+_\mu f)(p) \neq 0$, and a test function f' and a point p' such that every point of $\operatorname{supp} f'$ lies in the open future cone of p' and $(G^-_\mu f')(p') \neq 0$.
`Sealed.witness_kg_huygens_failure`

SHA-256 of `verify/Challenge.lean`: `a0f6e5043aa5beef82db4df564b695d7a04234464007411c60f26b2cc52e0da8`